

Andrew McGallian

Chicago, IL | 925-490-6049 | andymcgallian@gmail.com
github.com/amcgallian | linkedin.com/in/andrewmcgallian | amcgallian.github.io

Environmental scientist and geospatial developer taking projects from field collection to production software: GNSS and drone survey, laboratory and sensor work, Landsat pipelines, and deployed Python web applications.

EDUCATION

University of Chicago

Master of Science, Environmental Science | GPA 3.86

Bachelor of Science, Environmental Science; Minor in Geographic Information Science

Chicago, IL

Jun 2026

Jun 2025

EXPERIENCE

Geomatics Research Assistant

Türkmen-Karahöyük Archaeological Project, Institute for the Study of Ancient Cultures

Konya, Türkiye

Jun 2026 – Aug 2026

- Built and deployed the field team's QGIS plugin (below), cutting stratigraphic-unit polygon drawing from hours to tens of minutes by automating point-to-polygon generation from Emlid GNSS exports.
- Digitized 1,000+ stratigraphic units and flew ~20 drone surveys over 9 weeks across a 35-hectare site, producing orthomosaics, DEMs, and 200+ report-quality maps and phase plans.
- Designed ArcGIS Field Maps used daily by excavation teams and integrated GNSS, drone, and field data into project geodatabases with QA checks.
- Rebuilt the map layer of the project's archaeological records system in Vue and OpenLayers, used daily by excavators and lab teams to relate finds, samples, and trenches.

Data Visualization Fellow

University of Chicago Library, Center for Digital Scholarship

Chicago, IL

Sep 2025 – Jun 2026

- Built and deployed the Chicago Environmental Atlas, a containerized Flask/Docker app mapping 150 years of Chicago environmental change (1870–2025) over ~10 GB of Landsat tiles and vector layers behind a REST API.
- Built the analysis and visualization pipeline for a study of 32,014 UChicago dissertations (1893–2025), extracting research geography for 4,068 records via spaCy NER and geocoding.
- Designed and taught 3 quarterly workshops — ArcGIS Online, Survey123, and Python geocomputation — reaching 40+ participants, building the instructional datasets and documentation.

Undergraduate Research Assistant

Environmental Frontiers, Mansueto Institute for Urban Innovation

Chicago, IL

Jun 2024 – Jul 2025

- Led mobile street-level temperature sensing across Chicago neighborhoods, operating M-TEMP carts that recorded RTD temperature at 6 heights alongside GPS-tagged field surveys.
- Built an iOS app (SwiftUI, CoreLocation, SQLite) for real-time GPS-tagged field data logging.
- Processed the resulting data against Landsat 8/9 TIRS imagery, deriving emissivity-corrected land surface temperature to characterize urban heat exposure; presented at the American Geophysical Union Annual Meeting.

Undergraduate Research Assistant

Department of Geophysical Sciences, University of Chicago

Chicago, IL

Jun 2023 – Sep 2023

- Prepared and analyzed bulk carbonate samples by ICP-MS — powdering, dissolution, oxidative cleaning, and quality control — for an Early Eocene paleoclimate record.
- Analyzed trace metal datasets on carbonate diagenesis; presented at the Midwest Geobiology Conference.

PROJECTS

TKAP Tools — QGIS Plugin — Python, PyQGIS, GPL-2.0 | github.com/amcgallian/tkap-qgis

- Built and published a QGIS plugin used by an international archaeological field team, distributed via a self-hosted plugin repository so QGIS delivers updates automatically.
- Shipped three tools: GNSS points to stratigraphic-unit polygons, phase splitting with plan export, and section drawing over rectified wall photography.

No Evidence of Cloud Seeding Signatures in NEXRAD — MS Capstone — Anomaly detection, PyTorch | ENSC 36300, Spring 2026

- Built an unsupervised anomaly-detection framework (convolutional autoencoder + causal transformer) on NEXRAD radar to test whether cloud seeding operations (SNOWIE, Rainmaker) leave a detectable signature; transformer reached AUROC 0.96 on held-out test sequences.

TECHNICAL SKILLS

Languages, Tools & Visualization: Python (rasterio, GeoPandas, Pandas, NumPy, spaCy), SQL, JavaScript (ES6), R, Swift, Rust, Git/GitHub, Docker, Flask, DuckDB, PostGIS, Jupyter, Vega-Lite, Matplotlib, Blender

GIS & Remote Sensing: ArcGIS Pro, ArcGIS Online, Field Maps, Survey123, StoryMaps, QGIS (plugin development), Leaflet, OpenLayers, geodatabases, cartography, spatial analysis, remote sensing, photogrammetry

Field, Survey & Laboratory: GNSS/RTK (Emlid), drone survey, orthomosaic production, environmental monitoring and sampling, sensor deployment, ICP-MS, sample preparation, geochemical analysis